

H26EDU14

AI Placement Mentor

Personalized, AI-driven placement preparation for engineering students

TEAM LEAD

Monica Sherlin P

TEAM MEMBER

Rithuvarjitha VL

COLLEGE

VIT Chennai

Problem Statement & Objectives

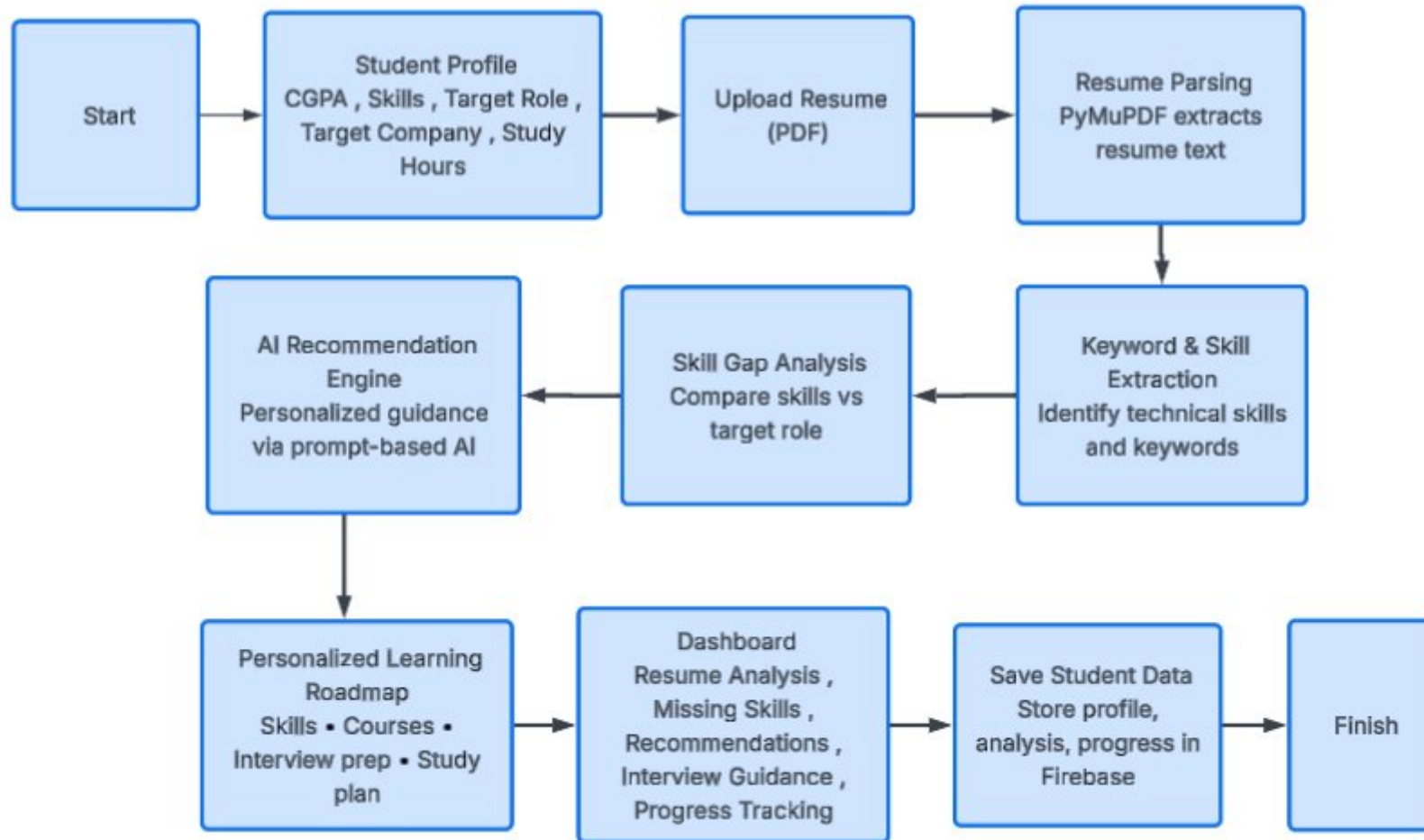
Problem

- No personalized guidance during preparation
- Weak resumes going unnoticed by students
- Difficulty identifying which skills to learn
- Preparation without a structured roadmap
- Low interview confidence
- Existing platforms offer generic resources

Our Objectives

- Analyze resumes automatically
- Detect missing skills for the target role
- Recommend what to learn next
- Generate a personalized learning roadmap
- Provide interview preparation guidance
- Track student progress over time

High-Level Architecture



Every module in this flow — parsing, keyword scanning, AI recommendation, and storage — is implemented and connected end to end.

Working Flow & Algorithms

Student Journey

- 1 Create profile & enter details
- 2 Upload resume (PDF)
- 3 AI analyzes resume & skills
- 4 Roadmap & recommendations generated
- 5 Results shown on dashboard

Algorithms Used

Resume Parsing

Extracts text from the uploaded PDF using PyMuPDF.

Keyword Matching

Detects technical keywords present in the resume.

Skill Gap Analysis

Compares current skills against target role requirements.

Prompt Engineering

Builds structured prompts from resume + profile data for personalized AI output.

BUILT WITH

Technology Stack

FRONTEND

- React.js
- Vite
- Tailwind CSS
- React Router
- Recharts

BACKEND

- Python
- Flask
- Flask-CORS

DATABASE

- Firebase

LIBRARIES

- PyMuPDF
- Firebase Admin SDK
- Requests
- python-dotenv
- Werkzeug

Results & Demo

Completed Features

- Student profile management
- Resume upload & parsing
- Skill extraction & gap detection
- Personalized AI recommendations
- Dashboard visualization

Demo Flow

Create profile → Upload resume → View skill gaps → See generated roadmap → Check dashboard

PlacementMentor

Prep season control center

You don't need more advice.
You need a **score**, a **gap list**,
and a plan.

Upload your resume once — get your readiness score, missing skills, a day-by-day roadmap, and a recruiter who tells you the truth.

v1.0 — hackathon build

New here

Resume session

Create your profile

Takes 30 seconds. This powers every score and recommendation you'll see.

FULL NAME

COLLEGE

YEAR

1st Year

CGPA

TARGET ROLE

SDE

TARGET COMPANY

Google

STUDY HOURS / DAY — 1H



CURRENT SKILLS (COMMA SEPARATED)

Create profile & continue

Comparison with Existing Solutions

Capability	Existing Solutions	AI Placement Mentor
Resume analysis	Not available	Implemented
Skill gap detection	Not available	Implemented
Personalized recommendations	Same for everyone	Based on resume & profile
Learning roadmap	Generic	AI-generated & personalized
Progress tracking	Not available	Implemented

SUMMARY

Summary & Future Work

AI Placement Mentor replaces generic placement content with a personalized roadmap, built from the student's own resume and goals.

FUTURE WORK

- AI Mock Interview
- ATS Score Prediction
- Coding Assessment Module
- Job Recommendation System
- Voice-based Interview Assistant
- Company-wise Preparation Plans
- LinkedIn Integration
- Mobile Application